

AIRCRAFT SIGNAL RECEIVER

Using RTL-SDR And DUMP1090

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ADS-B, or automatic dependent surveillance - broadcast, is used by commercial airplanes to broadcast their latitude and longitude along with other information to the air traffic controller. This information is transmitted at 1090MHz. ADS-B signals can be received using RTL-SDR dongle and decoded using software. In



Fig. 1: RTL-SDR dongle with antenna and remote

this project we use dump1090 program or command line to decode ADS-B signals on Linux machine. Apart from decoding ADS-B signals, dump1090 program also creates a Google Maps Web server where it displays the locations of airplanes whose ADS-B signals are being received.

Tools required for the setup

Tools required are laptop or computer running Linux/Ubuntu (preferably Ubuntu 14.04 and above), RTL-SDR dongle, antenna and dump1090 program installed on Ubuntu operating system (OS).

Installation of dump1090

Download dump1090 using the given link below.

<https://github.com/MalcolmRobb/dump1090>

Open the terminal and go to folder where dump1090 is saved. For installing it, type 'make' and press Enter, to install dump1090 on the computer.

Next, install rtl-sdr library by using the following command:

```
$ sudo apt-get install librtlsdr-dev
```

Getting started

To test the hardware, plug in RTL-SDR dongle to the Linux machine and run the following command on the terminal:

```
$ ./rtl_test -t
```

If everything is working fine, you will get an output similar to the following:

```
Found 1 device(s):
0: Realtek, RTL2838UHIDIR, SN: 00000001

Using device 0: Generic RTL2832U OEM
Found Rafael Micro R820T tuner
Supported gain values (29): 0.0 0.9 1.4 2.7 3.7 7.7 8.7 12.5 14.4 15.7 16.6
19.7 20.7 22.9 25.4 28.0 29.7 32.8 33.8 36.4 37.2 38.6 40.2 42.1 43.4 43.9 44.5
48.0 49.6
[R82XX] PLL not locked!
Sampling at 2048000 S/s.
No E4000 tuner found, aborting.
```

To use dump1090, refer to its manual for various commands and functions. For that, run the command given below.

```
$ ./dump1090 --help
```

A brief manual of dump1090 is given below.

dump1090 ModeS Receiver		Ver : 1.07.1908.13
--device-index <index>	Select RTL device (default: 0)	
--gain <db>	Set gain (default: max gain. Use -100 for auto-gain)	
--enable-agc	Enable the Automatic Gain Control (default: off)	
--freq <hz>	Set frequency (default: 1090 Mhz)	
--ifile <filename>	Read data from file (use '-' for stdin)	
--interactive	Interactive mode refreshing data on screen	
--interactive-rows <num>	Max number of rows in interactive mode (default: 15)	
--interactive-ttl <sec>	Remove from list if idle for <sec> (default: 60)	
--interactive-rtl1090	Display flight table in RTL1090 format	
--raw	Show only messages hex values	
--net	Enable networking	
--modeac	Enable decoding of SSR Modes 3/A & 3/C	
--net-beast	TCP raw output in Beast binary format	
--net-only	Enable just networking, no RTL device or file used	
--net-http-port <port>	HTTP server port (default: 8080)	
--net-ri-port <port>	TCP raw input listen port (default: 30001)	
--net-ro-port <port>	TCP raw output listen port (default: 30002)	
--net-sbs-port <port>	TCP BaseStation output listen port (default: 30003)	
--net-bi-port <port>	TCP Beast input listen port (default: 30004)	
--net-bo-port <port>	TCP Beast output listen port (default: 30005)	
--net-ro-size <size>	TCP raw output minimum size (default: 0)	
--net-ro-rate <rate>	TCP raw output memory flush rate (default: 0)	
--lat <latitude>	Reference/receiver latitude for surface posn (opt)	
--lon <longitude>	Reference/receiver longitude for surface posn (opt)	
--fix	Enable single-bits error correction using CRC	
--no-fix	Disable single-bits error correction using CRC	
--no-crc-check	Disable messages with broken CRC (discouraged)	
--phase-enhance	Enable phase enhancement	